

Task A (Implementing RLHF)

Train a reward model to predict human-like preference scores.

Steps:

1. Fine-tune a small transformer-based model, bert-base-uncased (<https://huggingface.co/google-bert/bert-base-uncased>), to predict the preference scores.
2. Train on the preference dataset using the Negative Log-Likelihood loss based on the Bradley-Terry model described below:
3. Save the trained reward model for later use.

Use Proximal Policy Optimization (PPO) to fine-tune a language model using the reward model.

Steps:

1. Load gpt2-medium (<https://huggingface.co/openai-community/gpt2-medium>) as the base language model.
2. Define a reinforcement learning loop where the model generates responses and the reward model assigns scores.
3. Use PPO to optimize the language model based on the reward scores and a reference model.

Use the frozen parameter version of the gpt2-medium (<https://huggingface.co/openai-community/gpt2-medium>) (i.e the frozen base model) as the reference model. You can use PPOTrainer from trl library.

4. Log the improvement in generated responses over training epochs.

Task B (Implementing DPO)

Use Direct Preference Optimization (DPO) to fine-tune the language model on the preference dataset.

Steps:

1. Load gpt2-medium (<https://huggingface.co/openai-community/gpt2-medium>) as the base language model.
2. Implement DPO loss function and optimize the model accordingly. Do not use DPOTrainer in this case.
3. Use the frozen version of gpt2-medium (<https://huggingface.co/openai-community/gpt2-medium>) as the reference model.
4. Log the improvement in generated responses over training epochs.

Task C (Performance Comparison)

Compare the performance of the 2 methods : RLHF and DPO

Steps:

1. Generate responses for the harmful or stereotype-triggering questions in the test split using the RLHF-trained and the DPO-trained models. Both of the models will be fine-tuned versions of gpt2-medium.
2. Evaluate them quantitatively (using automated metrics – BLEU and ROUGE). Use the “more_preferred” column from the dataset as the reference answer. You can use library functions for calculating BLEU and ROUGE scores.
3. Write a short report summarizing your findings, and do a contrast study of the 2 methods (RLHF and DPO) based on sample efficiency, response quality and computation cost.